



20 SQL QUERIES ON ONLINE BOOK STORE

MySQL Project – Data Analysis using SQL Queries



DATABASE SCHEMA

books

book_id (PK)
title
author
genre
price
stock
published_year

customers

customer_id (PK)
name
city
country

orders

order_id (PK)
customer_id (FK)
book_id (FK)
quantity
total_amount
order_date

- 1 Retrieve all books from fiction genre.

```
SELECT *  
FROM books  
WHERE genre = 'fiction';
```

- 2 Find books published after the year 1950.

```
SELECT *  
FROM books  
WHERE published_year > 1950;
```

- 3 List customers from Canada.

```
SELECT *  
FROM customers  
WHERE country = 'canada';
```

- 4 Orders placed in November 2023.

```
SELECT *  
FROM orders  
WHERE order_date BETWEEN  
'2023-11-01' AND '2023-11-30';
```

- 5 Total books available.

```
SELECT SUM(stock) AS total_count  
FROM books;
```

- 6 Details of most expensive books.

```
SELECT *  
FROM books  
ORDER BY price DESC  
LIMIT 1;
```

- 7 Customers who ordered more than 1 quantity of books.

```
SELECT *  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id  
WHERE o.quantity > 1;
```

- 8 Orders where total amount exceeds \$20.

```
SELECT *  
FROM orders  
WHERE total_amount > 20;
```

- 9 List all genre available in books.

```
SELECT DISTINCT genre  
FROM books;
```

- 10 Find the book with the lowest stock.

```
SELECT *  
FROM books  
ORDER BY stock ASC  
LIMIT 1;
```

- 11 Calculate the total revenue generate from the tables.

```
SELECT SUM(total_amount) AS total_rev  
FROM orders;
```

- 12 Total number of books sold for each genre.

```
SELECT b.genre, SUM(o.quantity) AS total_books  
FROM books b  
JOIN orders o ON b.book_id = o.book_id  
GROUP BY b.genre;
```

- 13 Find the average price of books in the fantasy genre.

```
SELECT AVG(price) AS average_price  
FROM books  
WHERE genre = 'fantasy';
```

- 14 Customers who have placed atleast 2 orders.

```
SELECT c.name, COUNT(o.order_id) AS order_count  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id  
GROUP BY c.name  
HAVING COUNT(o.order_id) >= 2;
```

- 15 Find the most frequently ordered book.

```
SELECT b.title, COUNT(o.order_id)  
FROM books b  
JOIN orders o ON b.book_id = o.book_id  
GROUP BY b.title  
ORDER BY COUNT(o.order_id) DESC  
LIMIT 1;
```

- 16 Top 3 most expensive books of fantasy genre.

```
SELECT title, price  
FROM books  
WHERE genre = 'fantasy'  
ORDER BY price DESC  
LIMIT 3;
```

- 17 Total quantity of books sold by each author.

```
SELECT b.author, SUM(o.quantity) AS total_quantity  
FROM books b  
JOIN orders o ON b.book_id = o.book_id  
GROUP BY b.author  
ORDER BY total_quantity DESC;
```

- 18 Cities where customers who spent over \$30 are located.

```
SELECT c.city, o.total_amount  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id  
WHERE o.total_amount > 30  
ORDER BY o.total_amount DESC;
```

- 19 Customer who spend more on orders.

```
SELECT c.customer_id, c.name,  
SUM(o.total_amount) AS total_spent  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id  
GROUP BY c.customer_id, c.name  
ORDER BY total_spent DESC  
LIMIT 1;
```

- 20 Calculate the stock remaining after fulfillment of all the orders.

```
SELECT b.book_id, b.title, b.stock,  
COALESCE(SUM(o.quantity), 0) AS total_quantity,  
b.stock - COALESCE(SUM(o.quantity), 0)  
AS remaining_quantity  
FROM books b  
LEFT JOIN orders o ON b.book_id = o.book_id  
GROUP BY b.book_id;
```

"SQL is not just a language, it's a way to communicate with data."

